Module 1:

1. What is data communication? List and explain characteristics and components of the communication model. (06 M)
2. Define switching. Explain Circuit Switched Network and Packet Switched Network. (06 M)
3. Explain the four basic topologies used in networks. List advantages and disadvantages of each of them.
4. What is meant by logical connection in TCP/IP. Explain with diagram how the identical objects interact.
5. With neat sketch, explain different layers of TCP/IP protocol suite. (08 M)
6. What are guided transmission media? Explain twisted pair cable in detail.
7. What is Virtual Circuit Network (VCN)? With neat diagram, explain three phases involved in VCN.
8. Write a note on encapsulation and decapsulation at Source Host for TCP/IP protocol suite.
9. Describe each layer of the TCP/IP and its responsibility.
10. Compare OSI and TCP/IP Models. What are the reasons for OSI model to fail?
11. Compare and contrast a Radio wave and microwave in terms of characteristics, range, propagation and applications.
12. Explain the concept of multiplexing & demultiplexing.

MODULE 2:

1. With a flow diagram and FSM, explain Stop and wait protocol.
2. Discuss HDLC common transfer modes.
3. Discuss the different types of HDLC frames.
4. A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces
   a. 1000 frames per second?
   b. 500 frames per second?
   c.250 frames per second?
5. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces
   a. 1000 frames per second.
   b. 500 frames per second.
   c. 250 frames per second.

6. Explain Pure ALOHA with procedure
7. With a neat diagram explain CSMA/CD
8. With a neat diagram explain CSMA/CA
9. Mention the different controlled access methods and explain any one in detail.
10. Define Redundancy. Explain CRC encoder and CRC decoder operation with block diagram. (08 M)
11. Distinguish between Flow Control and Error Control. Explain Stop and Wait Protocol. (08 M)
12. List and explain control fields of I-frames, S-frames, and U-frames. (04 M)
13. What is Hamming distance? With example, explain Parity Check Code.
14. Define Framing. Explain character-oriented framing and bit-oriented framing.
15. With flow diagram, explain CSMA/CA.
16. What is bit oriented framing and its frame pattern. Explain with example byte stuffing and unstuffing in bit oriented framing.
17. Explain the 3 different HDLC frames with diagram.
18. What is the difference between ALOHA and Slotted ALOHA
19. Explain CRC encoder and decoder for 4 bit dataword.
20. Explain stop and wait protocol with FSM and Flow diagram.
21. What is PPP? What are the services provided by PPP?

MODULE 3:

1. Discuss the services provided by Network Layer.
2. Explain packet switching datagram approach.
3. Explain packet switching virtual circuit approach.
4. Explain classfull addressing in detail.
5. Explain classless addressing in detail.
6. An organization is granted a block of addreses with beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses in its three subnets.
    a. One subblock with 10 addresses
    b. One subblock with 60 addresses
    c. One subblock with 120 addresses. Design the subblocks.
7. Explain DHCP message format.
8. Explain the operation of DHCP with a neat diagram.

9. With a neat diagram, explain the FSM of DHCP operation.
10. Explain the IPv6 datagram.
11. Comapre IPv4 and IPv6
12. With a diagram explain Least Cost Routing
13. Explain the distance vector routing algorithm with an example.
14. Explain the Link state routing protocol with an example.
15. Explain the path vector routing.
16. Explain the Operation of External BGP (eBGP)
17. Explain the Operation of Interenal BGP (iBGP)
18. Discuss the RIP message format
19. Discuss the performance of OSPF
20. Explain virtual-circuit approach to route the packets in packet-switched network. (10 M)
21. Illustrate the working of OSPF and BGP. (10 M)
22. Explain IPv6 datagram format. (10 M)
23. Write Dijkstra's algorithm to compute shortest path through graph. (06 M)
24. Write a note on Routing Information Protocol (RIP) algorithm. (04 M)
25. Explain classful addressing system with a neat diagram.
26. Write Dijkstra's algorithm to compute shortest path with an example.
27. Why is subnetting used and explain its importance.
28. Define and explain routing and forwarding in network layer.
29. Explain Open Shortest Path First Protocol with example.
30. Explain DHCP and its importance?

Module 4

1. Explain FSMs for the Go-Back-N protocol with a neat diagram.
2. Explain the TCP segment format
3. Illustrate connection establishment in TCP using Three-way handshaking using a suitable example.
4. Explain selective repeat protocol with its FSM.
5. Explain UDP services along with a neat diagram of pseudo-header for checksum.
6. Explain the Go-Back-N protocol along with sliding window diagrams.
7. Explain stop and wait ARQ protocol.
8. Explain TCP connection establishment and connection termination.
9. Explain the services provided by the transport layer to the upper layer.
10. With a neat diagram, explain the state transition diagram of TCP.

11. Explain connectionless and connection-oriented protocols in the transport layer.
12. Illustrate the state transition diagram of TCP
13. List and explain the features of TCP.
14. Illustrate the operation of Encapsulation and the decapsulation.
15. Explain Go-Back-N protocol working. **(10 M)**
16. With neat sketch, explain three-way handshaking of TCP connection establishment. **(10 M)**
17. With an outline, explain Selective Repeat protocol.
18. List and explain various services provided by User Datagram Protocol (UDP).
19. Draw the FSM diagrams for connectionless and connected oriented services offered by transport layer.
20. List the services and applications of UDP.
21. Explain Go-Bank-N protocol working.
22. Explain connection establishment of TCP using 3-way handshaking.
23. Explain TCP Congestion control.
24. Explain with example stop and wait protocol.

Module 5:

1. What are the services provided by application layer.
2. With the help of neat diagram , explain the client server paradigm.
3. With the help of neat diagram , explain the peer –peer paradigm.
4. Explain the process to process communication with the use of socket.
5. Illustrate on persistent connection in HTTP.
6. Illustrate on non persistent connection in HTTP.
7. Write the message formats of HTTP.
8. What are the functions of cookies
9. Explain the basic model of FTP.
10. Explain about electronic mail architecture.
11. Explain the concept of Local and Remote logging using TELNET
12. List and explain the components of SSH (Secure Shell).
13. Write a note on DNS
14. Briefly explain Secure Shell (SSH).
15. Write a note on Request and Response message formats of HTTP.
16. Explain Standard and Non-Standard Application Layer Protocols.
17. Differentiate client server paradigm and peer-to-peer paradigm.
18. Differentiate between Persistent and Non Persistent connection in HTTP.

19. Explain how data connections happens in File Transfer Protocol.
20. List the difference between local and remote logging.
21. Explain briefly Domain Name System (DNS)

Aspect	OSI Model	TCP/IP Model
Number of Layers	7 (Physical, Data Link, Network, Transport, Session, Presentation, Application)	4 (Network Access, Internet, Transport, Application)
Layer Mapping	- Physical + Data Link → Network Access - Network → Internet - Transport → Transport - Session + Presentation + Application → Application	- Network Access (Physical + Data Link) - Internet (Network) - Transport (Transport) - Application (Session + Presentation + Application)
Purpose	Theoretical framework for standardizing any network protocol.	Practical framework for internet-based communication.
Granularity	Highly granular with distinct Session and Presentation layers.	Less granular, merging Session, Presentation, and Application for simplicity.
Protocols	Protocol-agnostic; uses concepts like X.25, TP0-TP4 (not widely adopted).	Specific protocols: TCP, UDP, IP, HTTP, DNS, ICMP, ARP.
Adoption	Limited to education/reference; rarely implemented directly.	Widely implemented in the internet and real-world networks.
Development	By ISO, slow committee-driven process.	By DoD for ARPANET, rapid and community-driven.
Complexity	Complex due to seven layers and strict boundaries.	Simpler with four layers and overlapping functions.
Flexibility	Broadly applicable to any network type.	Focused on IP-based networks, less flexible but practical.
Reasons for OSI's Failure	- Complex, over-abstracted structure. - Lack of practical protocols. - Slow standardization. - Limited industry adoption. - Poor timing vs. TCP/IP's rise. - Inflexible for new technologies. - Perceived as bureaucratic.	N/A (dominant due to simplicity, robust protocols, early adoption, adaptability).